

AURÉLIEN GOUMAIN

Microelectronics Engineering Student

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PROFILE

Engineering student at Polytech Montpellier specializing in Microelectronics and Automation, with hands-on internship experience in analog IC design (Xschem, Magic, SkyWater 130nm PDK, Tiny Tapeout) and FPGA-based PCB design for radiation-testing research. Strong background in analog front-end design, embedded systems, and PCB development. Seeking an internship in IC design (analog, digital, or mixed-signal) or FPGA design to contribute to innovative hardware projects.

EDUCATION

Polytech Montpellier - MEng, Microelectronics and Automation (MEA) Sep 2024 - Present
Engineering school (1st year of engineering cycle), Montpellier, France

- Core coursework: analog and digital electronics, signal processing, automatic control, embedded systems, and programming.

CUPGE - University Paul Sabatier Sep 2022 - Jun 2024
2-year intensive preparatory program, Toulouse, France

- Advanced mathematics, physics, and engineering sciences. Admitted to Polytech engineering cycle on application.

Lycée Descartes - French Baccalauréat Sep 2019 - Jun 2022
Specialization: Mathematics, Physics & Chemistry, Montigny-le-Bretonneux, France

INTERSHIPS

Electronic Design Intern - University of Twente Apr 2026 - Present
Enschede, Netherlands. International research project

- Designing an FPGA-based PCB used to evaluate the radiation tolerance of the target FPGA component, within an international research environment.
- Led the technical research phase: selection of the FPGA under test, sensors, protection circuitry, and supporting modules to integrate on the board.
- Designed the full PCB architecture to host the FPGA and its peripherals under radiation-exposure conditions.
- Collaborating in English with an international team on research focused on electronic component reliability.

Analog IC Design Intern - LIRMM Feb 2026 - Mar 2026
Montpellier, France. Research laboratory

- Designed an analog integrated circuit using exclusively open-source EDA tools, as part of a project aimed at making IC fabrication more accessible.
- Produced the schematic and physical layout of the circuit using Xschem (design) and Magic (layout), targeting the open-source SkyWater 130nm PDK.
- Prepared the design for shared-silicon fabrication via Tiny Tapeout, reducing prototyping cost to ~500 EUR vs ~50,000 EUR with industry-standard proprietary flows.

ACADEMIC PROJECTS

ActInSpace Hackathon - "Coup de Cœur" Award Winner Jan 2026

- Took part in ActInSpace, an international hackathon organized by CNES and ESA, tackling real-world challenges faced by the space industry.
- Proposed reinventing the connectivity between modules inside a satellite by designing a unified connector standard, inspired by the USB-C approach, to simplify integration and improve modularity.
- Awarded the "Coup de Cœur" (Jury's Favorite) prize for the originality and potential impact of the concept.

Ultra-Low-Power Impedance Measurement System Polytech Montpellier

- Designed the analog front-end for a DAC-based impedance measurement circuit implemented on FPGA.
- Focused on low-power signal conditioning, impedance matching, and interface between the FPGA-generated digital signal and the analog measurement stage.

6-Axis Robotic Arm - Embedded Control System

Polytech Montpellier

- Controlled 6 servo motors using an STM32 microcontroller programmed in STM32CubeIDE.
- Developed low-level drivers through direct register manipulation for precise control of speed, torque, and closed-loop positioning.
- Implemented the forward and inverse kinematics for trajectory control of the arm.

Post-Stroke Actimetry Watch - Wearable Medical Device

Polytech Montpellier

- Developed a compact wearable device to monitor patient movement: accelerometer programming, display module integration, and power management.
- Performed PCB schematic capture, routing, and hand-soldering of the prototype.

Handwritten Digit Recognition - Machine Learning in MATLAB

Polytech Montpellier

- Designed, trained, and evaluated a neural network capable of classifying handwritten digits.
- Strengthened understanding of neural network architectures, data preprocessing, and performance analysis.

PROFESSIONAL EXPERIENCE

Private Science Tutor - Freelance

2022 - Present

Mathematics & Physics, middle and high school students

- Taught science with a focus on building student confidence and methodical problem-solving.
- Developed strong communication, pedagogy, and ability to explain complex concepts simply.

Assembly Line Operator - Pink Mobility

Jun 2024 - Jul 2024

Electric mobility manufacturer, summer job

- Worked on the electric scooter assembly line and handled stock management and order preparation.
- Gained first-hand exposure to hardware production processes and quality constraints.

Founder - E-Commerce Brand (Posters)

2023

Self-initiated entrepreneurial project

- Designed and launched a poster brand through a Shopify store: design, printing partners, and supplier coordination.
- Built and managed marketing campaigns on Google Ads, Meta (Facebook / Instagram), and TikTok; produced promotional video content.

TECHNICAL SKILLS

Programming: C, Python, MATLAB, assembly basics

Embedded Systems: STM32 (CubeIDE, register-level programming), FPGA development, ADC / DAC, SPI, I2C

Analog & Mixed-Signal: Signal processing, impedance matching, analog filter design, low-power front-end design

IC Design (Analog): Xschem (schematic capture), Magic (layout), SkyWater 130nm PDK, Tiny Tapeout flow, open-source EDA

PCB & EDA Tools: KiCad, Cadence, Quartus, Eclipse

Scientific Computing: MATLAB, Scilab

Other Tools: Linux, Git, Microsoft Excel

Languages: French (native), English (B2, professional working), Spanish (A1)

INTERESTS

Aviation & space exploration, automotive engineering, entrepreneurship & investing, philosophy & economics, sports.